

TARGET TRACKING GIMBAL

Active Gimbal Control

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Abstract Many ISR, military, and law enforcement applications require pointing a sensing payload at a moving target. Aircraft can be fitted with a sophisticated sensor suite consisting of Electro-Optical/InfraRed (EO/IR) instruments in combination with a laser rangefinder (LRF), or other remote sensing systems and combinations on a two-axis gimbal to enable precise target tracking. This note provides a detailed analysis for choosing the right GNSS/INS system for such applications, alongside other integration considerations that will determine total system performance. The error budget is derived and an example provided to demonstrate the impact of various design choices.

1 INTRODUCTION

Gimballed systems are one of the most common payloads on everything from UAVs to marine vessels, enabling pointing of that payload independent of the vehicle attitude. That independent motion also allows for stabilization, decoupling the high-frequency motion and vibrations of the platform or vehicle from those of the payload. Gimballed payloads include everything from cameras or other remote sensing systems to communication antennas, and are used for a variety of applications.

Many applications require a specific target to be followed or tracked from an aerial vehicle. Such target tracking applications typically incorporate a two-axis gimbal pointing an Electro-Optical/InfraRed (EO/IR) imager combined with a laser rangefinder (LRF) to provide geo-referencing for a tracked target. Real-time processing of the data from the EO/IR imager allows for image-based target identification and tracking, which is often used to steer the gimbal so as not to require human steering. This application note focuses on such target tracking applications, but the following application notes provide additional resources for other gimbal applications:

- AN201: Gimbal Stabilization and Pointing
- AN202: Gimballed EO/IR Systems
- AN203: Satellite Communications (SATCOM)

While single-axis gimbals are used occasionally, most systems utilize two-axis gimbals to provide full pointing control and will be the focus of this note. Two-axis gimbals are sometimes referred to as az-el stages or pan-tilt units and can be broadly broken down into three distinct components: (a) the base, which is fixed to whatever platform or vehicle is carrying the gimbal, (b) the outer axis, which moves relative to the base, typically the pan or azimuth axis, and (c) the inner axis, which moves relative to the outer axis and provides tilt or elevation control. The remote sensing system is mounted to the inner axis.

Integrating the GNSS/INS system with the vehicle and gimbal can be approached many different ways, as described in AN201. For this particular application, we will assume the most accurate approach has been taken, which is to incorporate one GNSS/INS onto the vehicle

or the base of the gimbal, while a second GNSS/INS system is incorporated onto the inner axis of the gimbal with the remote sensing payload.

2 SENSOR SELECTION & UTILIZATION

Gimballed systems typically require two separate forms of inertial navigation feedback: (a) gyro stabilization to correct for high-frequency motion of the payload and (b) absolute pointing feedback for geo-referencing and target tracking.

2.1 Selection

In most gimbal pointing applications, including target tracking, the gimbal control system requires position data, which precludes a pure IMU or AHRS solution and requires a GNSS-aided option. Even for those applications where positioning is not required, the high-accuracy pointing requirements during dynamic motion means that the degraded performance an AHRS suffers during such dynamics is unacceptable and a GNSS-aided solution is required.

Keeping eyes on target requires both gyro stabilization and absolute pointing. Placing a GNSS/INS on both the inner axis and the base of the gimbal provides the highest absolute pointing accuracy for geo-referencing, as well as direct angular rate feedback for the PID control of the gimbal. For applications with active control requirements, a real-time, low-latency capability is required for the navigation system.

2.2 Setup for Two GNSS/INS

For this application, it is assumed that one GNSS/INS is mounted onto the vehicle or base of the gimbal and a second GNSS/INS system is incorporated onto the inner axis of the gimbal. In this approach, having the first GNSS/INS mounted on the base/platform allows the GNSS antennas to be mounted at any convenient location on the platform. Rather than using the orientation outputs, however, the first GNSS/INS provides the position & velocity of the center of the gimbal axis. In this case, a dual-antenna GNSS/INS does not provide any additional benefit and a single-antenna GNSS/INS is typically employed instead.

Rather than rely on GNSS measurements from its own antenna, the second GNSS/INS system mounted at the center of the gimbal inner-axis uses the output from the first GNSS/INS system as pseudo-GNSS measurements, as if it had a GNSS antenna mounted at that location. Having an INS mounted directly on the inner-axis provides the highest performance possible, while allowing the GNSS antennas to be mounted anywhere on the platform. Since this second GNSS/INS uses pseudo-measurements from the first GNSS/INS receiver rather than using measurements from its own onboard GNSS receiver, a single-antenna GNSS/INS is generally used. Often this approach utilizes an industrial-grade GNSS/INS on the base, with a tactical grade or higher-end INS mounted to the inner-axis for improved stabilization and pointing.

3 GEO-REFERENCING

In order to track a target it is often necessary to geo-reference a point or a feature in an image to a global coordinate frame. The basis of this solution lies in the geo-referencing equation (Eq. 1), which can be used to solve for the point on ground (p_g) of the image feature.

$$p_g = p_a + L[C]\hat{\rho} \quad (1)$$

Where p_a is the estimated aircraft position, $[C]$ is the estimated direction cosine matrix mapping from the camera to inertial coordinate frame, $\hat{\rho}$ is the normalized camera feature vector, and L is the slant range between the gimbal and the point on the ground, as seen in Figure 1.

The estimated aircraft position vector (p_a) comes directly from the GNSS/INS system and is subject to both GNSS position errors and timing errors between systems. The coordinate frame transform between the camera's reference frame and an inertial coordinate frame (eg. NED) is a combination of the attitude measured by the GNSS/INS and the transformation between the camera and INS as measured by the gimbal encoders. Errors in $[C]$ come primarily from sensor attitude misalignments, GNSS/INS attitude uncertainty, and timing errors between sensors.

The camera feature vector (ρ) is defined in Eq. 2, where f is the focal length of the camera and (x, y) is the pixel location of the image feature. Errors in $\hat{\rho}$ are created by camera feature detection errors, lens distortions, and errors in the focal length calibration.

$$\rho = [x \ y \ f]^T \quad (2)$$

The slant range to the target (L) can come directly from a laser range-finder or similar sensor. Errors in L are directly determined by the uncertainty in the sensor's measurements.

Aerial Geo-Referencing

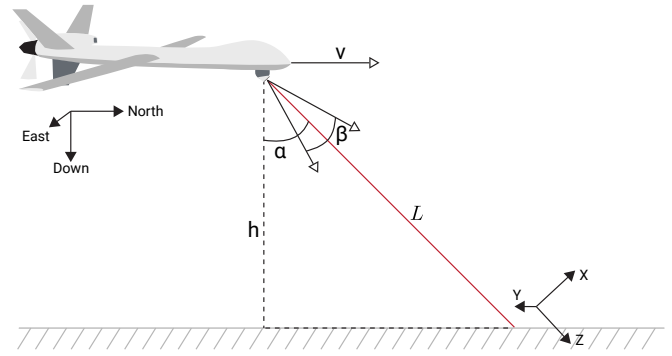


FIGURE 1

4 GNSS/INS ERROR CONTRIBUTIONS

The terms p_a and $[C]$ in Eq. 1 derive from the measurements of the GNSS/INS system. The uncertainty associated with those measurements is typically represented by a standard deviation (σ). The measurements can be split into position and attitude measurements, each having their own effect on the error budget.

4.1 Position Errors

Position errors come from the uncertainty in position of the GNSS/INS sensor. This position uncertainty is represented in an inertial North, East, Down (NED) coordinate frame. The position uncertainty of a GNSS/INS is mostly determined by the accuracy of the GNSS solution alone, and due to the nature of GNSS, typically has double the uncertainty vertically as horizontally.

While standard single-frequency GNSS can provide a position estimate with around 2 m horizontal uncertainties, errors can be reduced substantially by switching to a real-time kinematic (RTK) or post-processed kinematic (PPK) multi-frequency GNSS receiver, which yields uncertainties in the centimeter range. These require real-time corrections transmitted from an independent base station or time to post-process such corrections after logging considerable raw data from the GNSS, respectively. For more information on RTK and PPK, see the online VectorNav Library. Table 1 provides typical performance values for the different GNSS options and the axes along which the error projects.

GNSS Position Error Values

ERROR SOURCE	AXES	STANDARD GNSS	RTK GNSS
Horizontal position (σ_h)	N, E	2 m	0.02 m
Vertical position (σ_v)	D	5 m	0.05 m

TABLE 1

4.2 Attitude Errors

Any angular uncertainty represented by σ_a [deg] projects to a positional error, $\bar{\sigma}_a$ [m], based on some distance, d [m], as seen in Eq. 3. The distance d is typically a function of the slant range, L .

$$\bar{\sigma}_a = d \frac{\sigma_a \pi}{180} [m] \quad (3)$$

For GNSS/INS attitude uncertainty, the exact projection from yaw-pitch-roll uncertainties to position errors on the ground are found using Eq. 3 and the values represented in Table 2. Note that the values of α represents the look-angle off-vertical of the gimballed EO/IR sensor, with zero being straight down and 90° being horizontal.

INS Attitude Error Projections

ERROR PROJECTION	σ_a	d
Yaw-Position ($\bar{\sigma}_\psi$)	σ_ψ	$L \sin \alpha$
Pitch-Position ($\bar{\sigma}_\theta$)	σ_θ	L
Roll-Position ($\bar{\sigma}_\phi$)	σ_ϕ	$L \cos \alpha$

TABLE 2

A system designer typically must decide between two different INS performance grades that will impact the attitude uncertainty error contribution: industrial grade and tactical grade. Representative performance values for each can be found in Table 3, along with the axes along which the errors project.

INS Attitude Error Values

ERROR SOURCE	AXES	INDUSTRIAL	TACTICAL
Yaw (σ_ψ)	y	0.3°	0.1°
Pitch (σ_θ)	x	0.1°	0.03°
Roll (σ_ϕ)	y	0.1°	0.03°

TABLE 3

4.3 Total Errors

Combining uncertainties is accomplished via a sum of squares technique, as shown in Eq. 4 for combining the attitude and positional uncertainties from a GNSS/INS.

$$\bar{\sigma}_{\text{TOT}} = \sqrt{\bar{\sigma}_p^2 + \bar{\sigma}_a^2} \quad (4)$$

Figures 2 show the geo-referencing uncertainties due only to the GNSS/INS when directly overhead (gimbal pointing straight down), depending on industrial-grade vs tactical-grade inertial performance, and standard L1-only GNSS vs. RTK GNSS positioning. At short ranges (<100m), the GNSS accuracy dominates the system accuracy, whereas at long ranges (>1000m) the INS performance is determinative.

Projected GNSS/INS Errors ($\alpha = 0^\circ$)

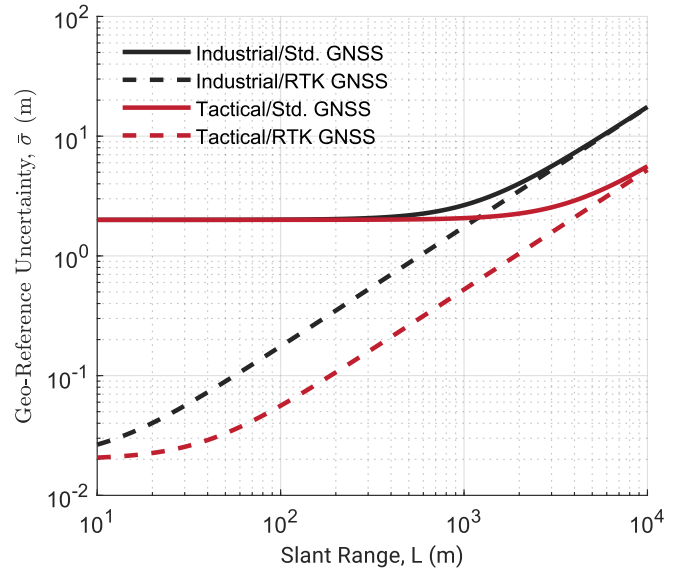


FIGURE 2

5 INTEGRATION ERRORS

The GNSS/INS uncertainties are not the only sources of error in the geo-referencing problem. How the GNSS/INS is integrated with the rest of the systems and the capabilities of the EO/IR and LRF systems can introduce errors larger than the GNSS/INS errors if care is not taken.

5.1 Misalignment Errors

Misalignment error is an orthogonality error caused by the sensing axis not aligning perfectly to the desired input axis. In this case, misalignment errors are due to the misalignment between the EO/IR imager and the navigation system, causing a mismatch between the GNSS/INS attitude and the attitude of the payload. These misalignment errors can be broken down into a variety of sources:

- **GNSS/INS Axes to GNSS/INS Case** A standard INS specification is the post-calibration misalignment between the actual sensing axes of the inertial sensors and the case in which they are mounted. This misalignment is typically small, but it is not negligible in long-range pointing applications.
- **GNSS/INS Case to Imager Case** The misalignment between the GNSS/INS case and the EO/IR imager case is minimized under the approach described here, since the inner-axis GNSS/INS can be mounted directly to the camera. This minimizes the tolerance stack-up errors, but alignment accuracy still depends on the manufacturing tolerances of that interface and the use of alignment pins.
- **Imager Axes to Imager Case** Similar to the GNSS/INS, there will always be residual errors in the alignment of the imager to the camera case. These errors can be quite large on an uncalibrated camera, especially those without precision mounting surfaces.

Calibration and Mitigation

While each of the misalignment error sources listed could be calibrated independently, the best approach to mitigating misalignment errors is to perform a calibration mapping the internal electronics of the payload (EO/IR imager) directly to the internal electronics of the GNSS/INS. The exact procedure for this varies depending on the nature of the payload sensor and the specifics of the application, but VectorNav has extensive experience in this area and has a variety of procedures and tools available to support these calibrations.

Error Budget Contribution

Regardless of the source of the misalignment, the errors can be modeled as an attitude error similar to the GNSS/INS attitude error. The total effect of misalignment errors on the error budget can be calculated using Eq. 3, with $\sigma_a = \sigma_{MA}$, and $d = L$. These errors are always perpendicular to the line-of-sight. When properly calibrated, total misalignment error can be as low as 0.01° .

5.2 EO/IR and LRF Sensor Errors

Errors in the EO/IR sensor result from either errors in the intrinsic parameters of the sensor, or based off of the feature detection and correlation. The three major intrinsic errors are distortion, zero-offset error, and focal length error. Distortion is a major intrinsic error coming from the camera lens which causes straight lines to appear as curved. Meanwhile, zero-offset errors (an inaccurate center of the image) and focal length errors lead to incorrect direction vectors. Feature detection errors result from the limited resolution of the image as well as the precision of the feature detection algorithm. Errors coming from the LRF directly come from the uncertainty in the sensor's measurement of slant range.

Calibration and Mitigation

While the intrinsic properties of the camera may be calibrated in a lab environment, the distortion in the camera changes with the focal length, which may be much longer for aerial surveying applications than can be tested in a lab. As such, the intrinsic errors have to be calibrated in-situ by using ground reference points and special calibration software. Assuming high-performance image processing algorithms, the resolution of the camera is the main determinant of feature tracking errors.

Error Budget Contribution

The error budget contribution from the camera (σ_{EO}) can be characterized by a feature location accuracy (δ [px]), which is typically a fraction of a pixel and encompasses feature detection errors and residual intrinsic parameter calibration errors. Given the resolution (R [px]) and the field-of-view (β [°]) of the imager, the error budget contribution can be calculated as an attitude error using Eq. 3, with $\sigma_a = \beta\delta/R$ and $d = L$. These errors are always perpendicular to the line-of-sight, in the x- & y-axes. The error budget contribution from the LRF uncertainty is parallel to the line of sight in the z-axis.

5.3 Timing Errors

Timing errors in an targeting system are caused by an offset in time between when a payload measurement is taken and when GNSS/INS data is captured. This offset in time is due to delays in the payload sensor, delays in the GNSS/INS, and delays in the computing platform used to collect both sets of data. These delays can be broken down into static bias (b_t) and a random error known as jitter (σ_t). When not properly time aligned, errors arise in both position and attitude due to the product of the linear or angular velocity and the time error.

Calibration and Mitigation

The best course of action is to utilize synchronization or timing signals, such as a SyncIn or SyncOut signal on VectorNav products, which allows for microsecond level timestamping of the time of validity of each measurement. Alternatively, a calibration can be performed on the gimballed system to estimate the static bias portion of the timing error, though the jitter errors will remain larger than when using a synchronization signal. This calibration is best performed in situ, as the timing errors can be hard to measure without motion. This typically involves imaging a surveyed-in point on the ground on at least two different passes at different speeds, as timing errors are indistinguishable from misalignment errors at a single, fixed velocity.

Error Budget Contribution

The error budget contribution of the timing errors is dependent on the angular velocity (ω) and linear velocity (v) of the vehicle, and can be characterized by Eq. 5. The errors due to linear velocity are in the direction parallel to the velocity of the vehicle, while errors due to angular velocity are in the direction perpendicular to the angular velocity vector.

$$\bar{\sigma}_t = \left(v + \frac{L\omega\pi}{180} \right) (b_t + \sigma_t) \quad (5)$$

On an uncalibrated system, the static bias can be on the order of tens of milliseconds, whereas a well-calibrated system eliminates the static bias and leaves a jitter on the order of a few milliseconds, and a system using synchronization signals achieves jitter on the order of microseconds.

Latency

In most target tracking applications, the GNSS/INS provides real-time feedback to control the gimbal in addition to its geo-referencing function. The latency of the GNSS/INS - the time between real-world motion and the GNSS/INS outputting measurements of that motion - impacts gimbal stabilization and active target tracking. VectorNav's engineers can provide a detailed assessment of latencies across our sensors, all within a few milliseconds.

6 EXAMPLE

To illustrate how these different error terms contribute to an overall targeting application error budget, we consider the example of an aircraft flying flat and level, due north with the gimbal tracking a point on the ground along the north-south line of flight. We assume that both the misalignments and system timing have been well calibrated, and that the gimbal is fixed at the time (zero angular velocity). The parameters used to define the mission and system errors are given in Table 4, while the parameters evaluated for the GNSS/INS are given in Tables 1 & 3.

Example Parameters

DESCRIPTION	SYMBOL	VALUE
Camera FoV	γ	30°
LRF accuracy	σ_{LRF}	1 m
Camera resolution	R	1080 px
Distortion model accuracy	δ	0.5 px
Height from ground	h	1000 m
Velocity of the vehicle	v	50 m/s
Misalignment errors	σ_{MA}	0.01°
Timing jitter	σ_t	5 ms

TABLE 4

6.1 NED Frame and Slant-Range Frame

When discussing the error budget contributions it is important to decompose error budgets into an inertial coordinate frame and a relative slant range frame. An inertial coordinate frame, such as North-East-Down (NED) is independent of the orientation of the gimbal, while the relative slant range frame shown in Figure 3 rotates with the line-of-sight to the target.

In the special case of $\alpha = 0$, the slant-range frame aligns with the NED frame and the total error can be easily summed. Contact VectorNav for information on combining errors from the NED and slant-range frames in the more general case of $\alpha \neq 0$.

6.2 Error Sources

Each of the error sources described earlier must be accounted for to determine total system performance. Keep in mind that the simplified error models used here are useful for design purposes, but empirical field testing is required to evaluate true system performance.

GNSS/INS Position Errors

The error budget from horizontal GNSS/INS uncertainty directly comes from the GNSS horizontal position uncertainty in an NED coordinate frame. For standard GNSS, this gives an error of 2 m in the North and East axes and 5 m in the Down Axis.

Projecting Error Onto Target Position

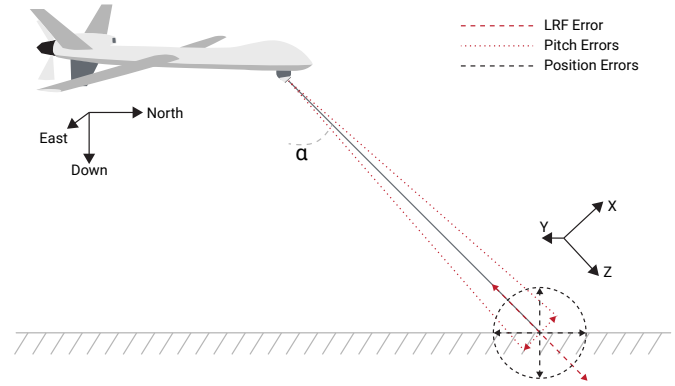


FIGURE 3

Attitude Errors

Position uncertainties resulting from attitude errors include those from the GNSS/INS, misalignment errors, and EO/IR errors. These positional errors have been given with respect to the slant-range frame, with the xy-plane perpendicular to the line-of-sight.

Laser Rangefinder Errors

The error contribution from the LRF comes from the uncertainty of the slant range being measured from the LRF. This error contributes directly the z-axis error parallel to the slant range. The LRF error contribution is also the only direct contributor to the z-axis error budget.

Timing Errors

The error budget contribution from time is caused by the time delay in the measurements. Since we assume the gimbal is not changing attitudes only linear velocity needs to be compensated for and Eq. 5 reduces to:

$$\tilde{\sigma}_t = v\sigma_t \quad (6)$$

The error contributed by the timing is in the direction of the velocity of the vehicle, which for this example is due North on the NED axis. Keep in mind that utilizing a dedicated synchronization signal can reduce the timing error contribution by an order of magnitude or more.

6.3 Calculated Results

The results of this simple example are broken up by individual error contributions in the inertial NED frame and the slant-range frame in Tables 5 & 6, respectively. This includes look-angles of 0°, and 60° – at which point the slant range is twice the altitude.

The total error is plotted in Figures 4 & 5 to show the North-East and North-Down axes of error. The plotted error is calculated for the special case when the gimbal look angle is 0° and the slant-range frame axes align with the NED frame. Despite the simplifications of this example, these provide a reasonable estimate of the errors experienced by a targeting system.

Position Errors: NED Frame

ERROR	OPTION	AXIS	$\alpha = 0^\circ$	$\alpha = 60^\circ$
$\tilde{\sigma}_h$	Std. GNSS	N, E	2.0 m	2.0 m
$\tilde{\sigma}_v$	Std. GNSS	D	5.0 m	5.0 m
$\tilde{\sigma}_h$	RTK GNSS	N, E	0.02 m	0.02 m
$\tilde{\sigma}_v$	RTK GNSS	D	0.05 m	0.05 m
$\tilde{\sigma}_t$	All	N	0.25 m	0.25 m

TABLE 5

Position Errors: Slant-Range Frame

ERROR	OPTION	AXIS	$\alpha = 0^\circ$	$\alpha = 60^\circ$
$\tilde{\sigma}_\psi$	Industrial	y	0.0 m	9.0 m
$\tilde{\sigma}_\theta$	Industrial	x	1.75 m	3.5 m
$\tilde{\sigma}_\phi$	Industrial	y	1.75 m	1.75 m
$\tilde{\sigma}_\psi$	Tactical	y	0.0 m	3.0 m
$\tilde{\sigma}_\theta$	Tactical	x	0.52 m	1.0 m
$\tilde{\sigma}_\phi$	Tactical	y	0.52 m	0.52 m
$\tilde{\sigma}_{MA}$	All	x, y	0.18 m	0.35 m
$\tilde{\sigma}_{LRF}$	All	z	1.0 m	1.0 m

TABLE 6

Total Error In North-East Axes ($\alpha = 0^\circ$)

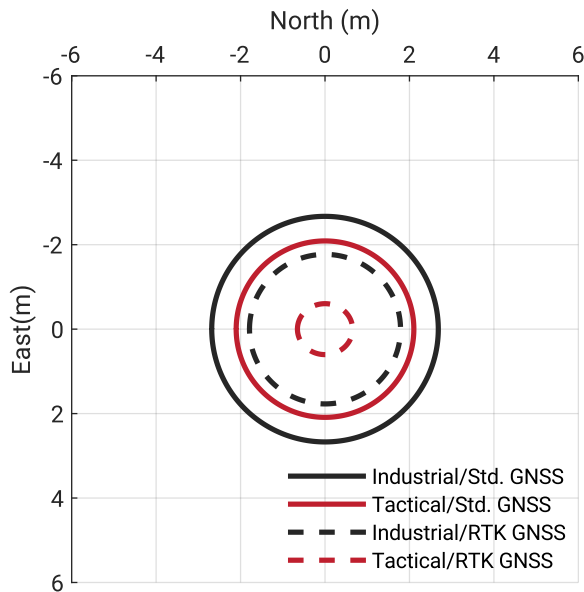


FIGURE 4

Total Error In North-Down Axes ($\alpha = 0^\circ$)

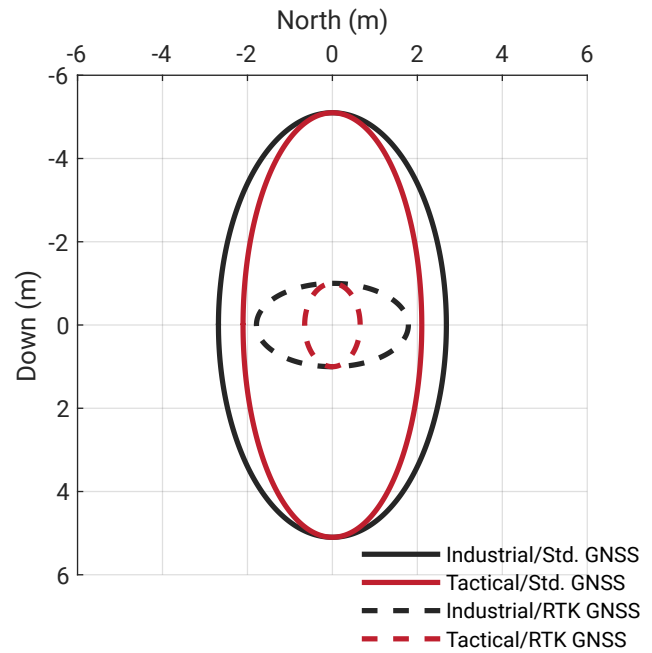


FIGURE 5

7 CONCLUSION

Some aerial applications require a point or feature to be followed in real-time, referred to as target tracking applications. For many of these applications an EO/IR camera and an LRF is mounted on a gimbal to find and geo-reference points of interest. Gimbals typically need their own GNSS/INS, independent of the aircraft navigation system, and for best results utilize two GNSS/INS systems mounted to the base of the gimbal and to its innermost axis.

While low GNSS/INS uncertainties can be achieved with a VectorNav Industrial Series GNSS/INS, uncertainties can be reduced substantially by using RTK-enabled GNSS or a VectorNav Tactical Series GNSS/INS, or both. Misalignment errors between the INS and the camera, camera errors like lens distortions, and time synchronization errors between the two systems can each contribute significantly to the total error budget. Each of these errors can usually be calibrated out and become mostly negligible in the system.

If you would like to learn more about the recommended solution for your particular application, the impacts that various design choices may have, or recommendations on calibration procedures, please contact VectorNav for a more in-depth discussion.



VectorNav Technologies, LLC
10501 Markison Rd
Dallas, TX 75238, USA

Tel: +1.512.772.3615
Email: sales@vectornav.com
Web: vectornav.com

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